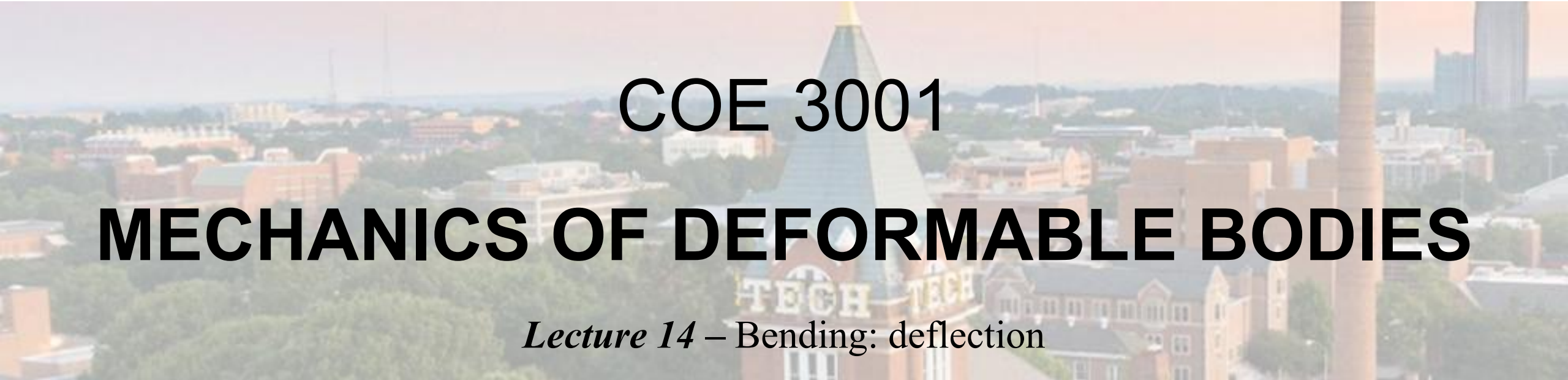




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A wide-angle, slightly hazy photograph of the Georgia Institute of Technology campus. The image shows various red-brick buildings, green trees, and a tall smokestack in the distance. The text is overlaid on this image.

COE 3001

MECHANICS OF DEFORMABLE BODIES

Lecture 14 – Bending: deflection

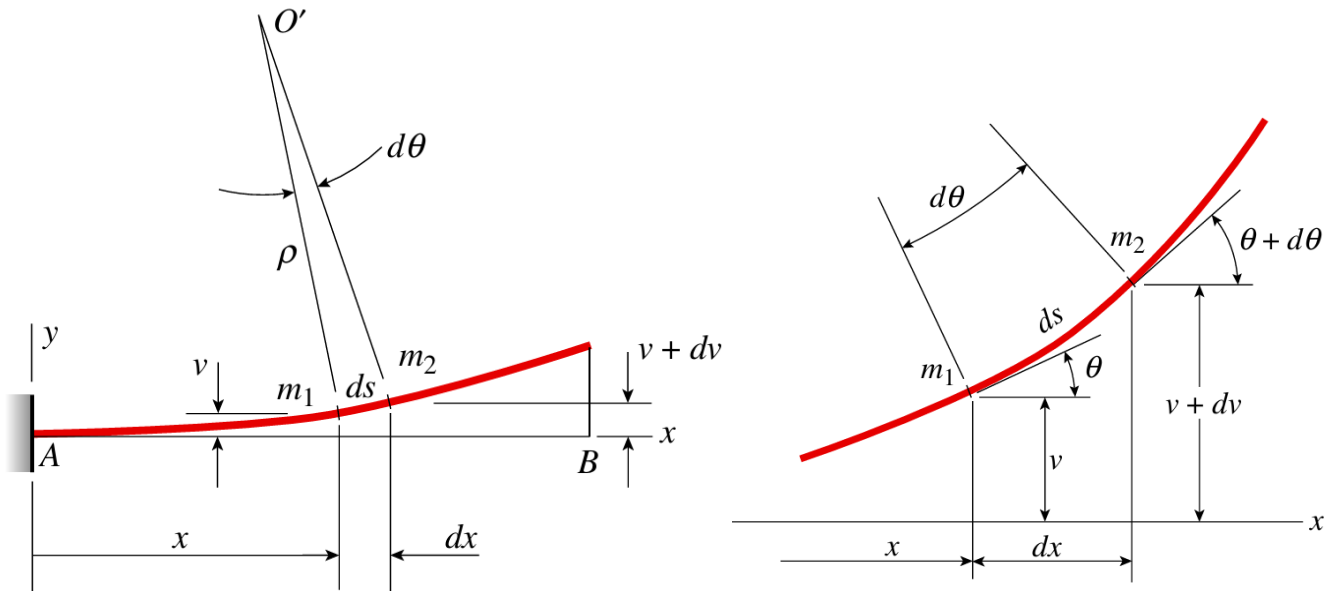
Yazhuo Liu

Georgia Institute of Technology

Mar. 11, 2026

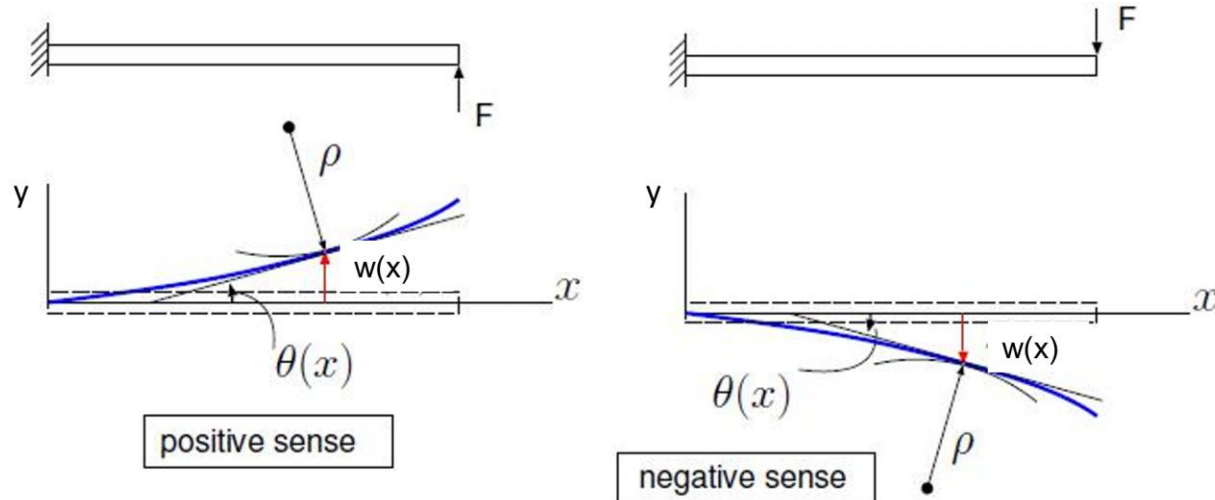
Bending deflection

When a beam with a straight longitudinal axis is loaded by lateral forces, the axis is deformed into a **curve**, called the **deflection curve** of the beam.



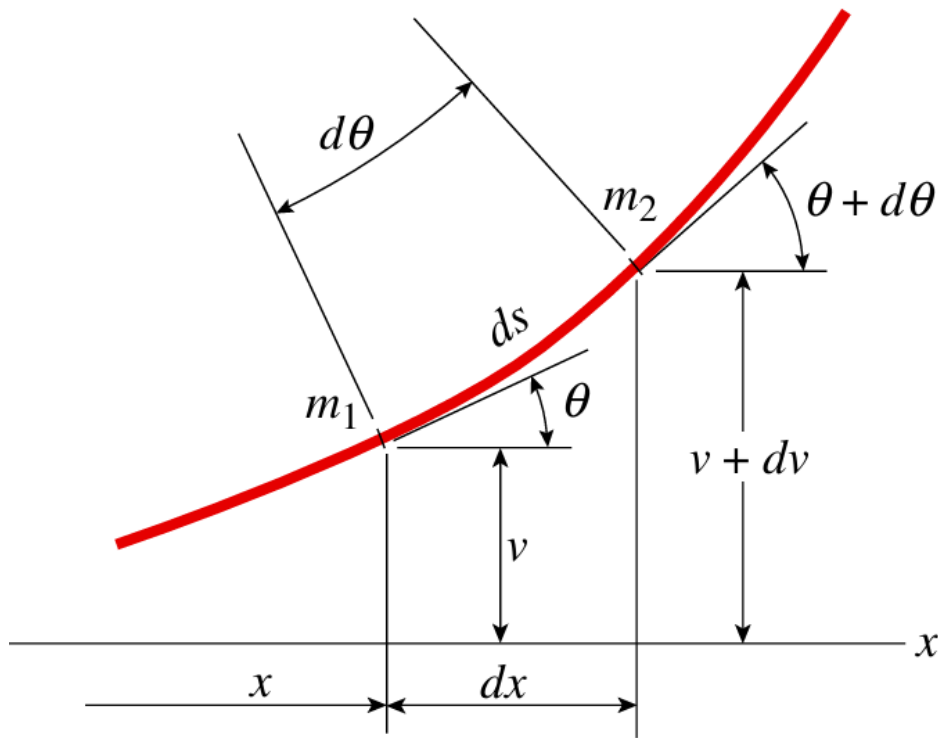
- Deflection (v): the vertical displacement of the centroid of cross-section
- Deflection angle, also called angle of rotation (θ): the angle between the x axis and the tangent to the deflection curve

Sign convention



- Because the y axis is **positive upward**, the deflections are also positive when upward
- For the choice of axes (x positive to the right and y positive upward), the angle of rotation is positive when **counterclockwise**.

Deflection angle and deflection



Geometrically,

$$\rho d\theta = ds \Rightarrow \kappa = \frac{1}{\rho} = \frac{d\theta}{ds}$$

$$\tan \theta = \frac{dv}{dx} \Rightarrow \theta = \arctan \frac{dv}{dx} = \tan^{-1} \frac{dv}{dx}$$

Under small strain assumption,

$$ds \approx dx \Rightarrow \kappa = \frac{d\theta}{dx}$$

$$\theta \approx \tan \theta = \frac{dv}{dx}$$

Then

Recall: $\kappa = \frac{M}{EI}$

$$\kappa = \frac{d^2v}{dx^2}$$

Differential equation of the deflection curve

$$\frac{M}{EI} = \kappa = \frac{d^2 v}{dx^2}$$

For prismatic beam (constant EI):

$$EI \frac{d^2 v}{dx^2} = M$$

Bending-moment equation

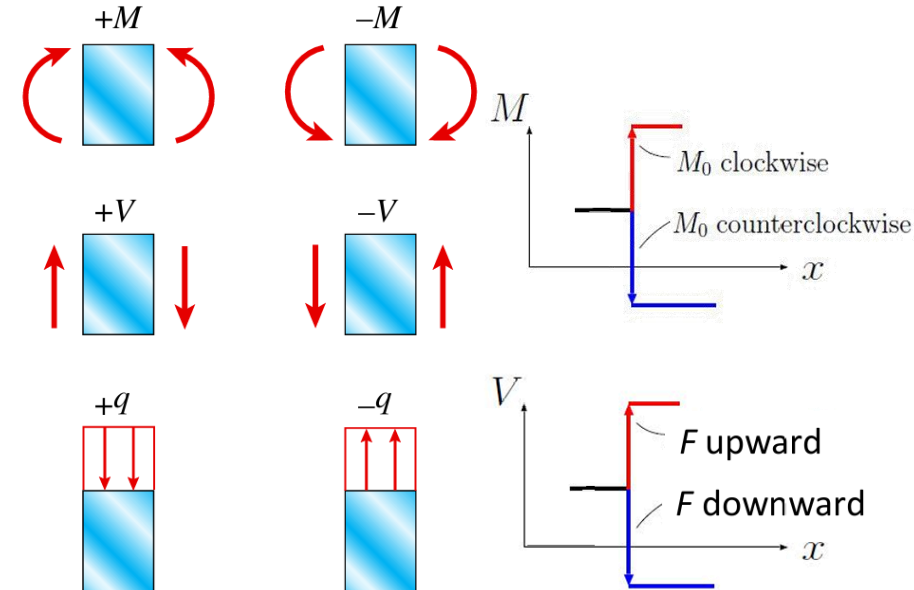
$$V = \frac{dM}{dx} = EI \frac{d^3 v}{dx^3}$$

Shear-force equation

$$-q = \frac{dV}{dx} = EI \frac{d^4 v}{dx^4}$$

Load equation

Sign conventions for bending moment M , shear force V , and intensity q of distributed load



Bending-moment equation

Bending-moment equation

$$EI \frac{d^2 v}{dx^2} = M$$

Integration once, we can know deflection angle

$$EI\theta \approx EI \frac{dv}{dx} = \int_0^x M(x) dx + C_1$$

Integration again, we can know deflection equation

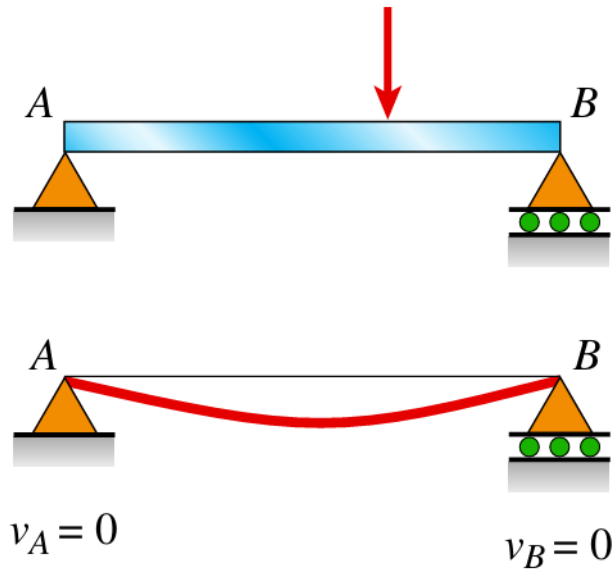
$$EIv = \int_0^x \int_0^x M(x) dx dx + C_1 x + C_2$$

Integration constant C_1 , C_2 can be known by boundary conditions of beam deflection curve and continuity of deformation conditions.

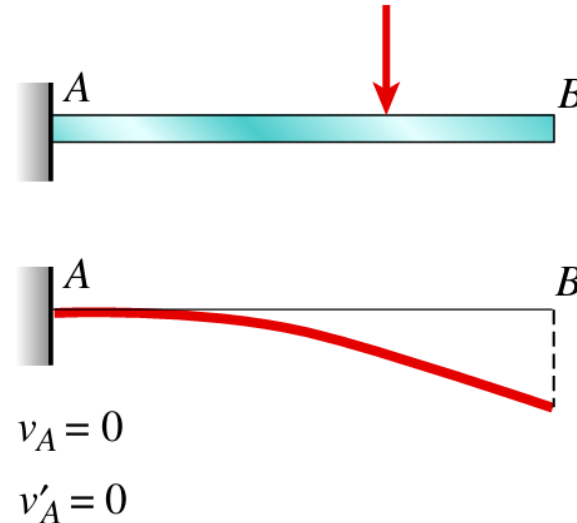
Method of
successive
integrations

Boundary and continuity conditions

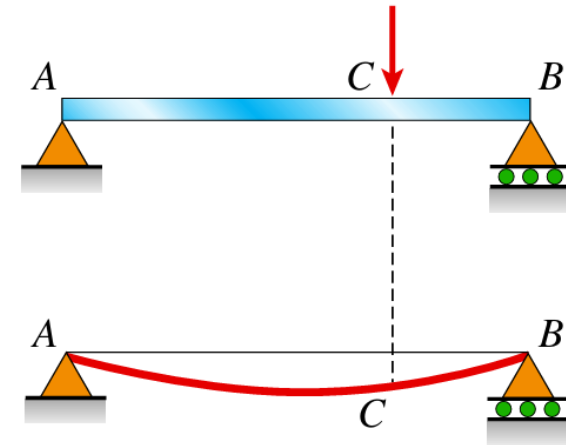
Boundary conditions at simple supports



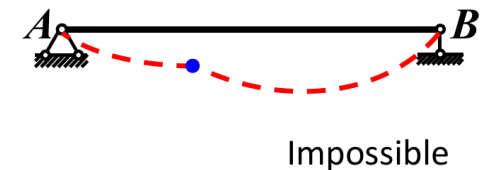
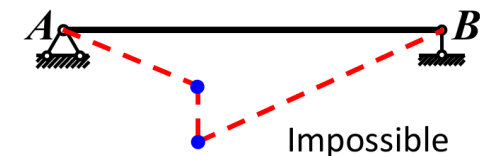
Boundary conditions at a fixed support



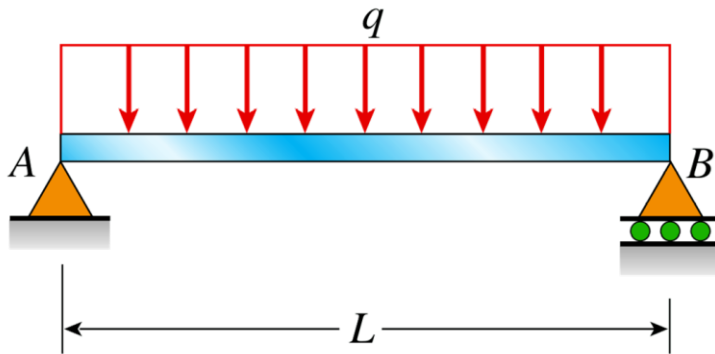
Continuity conditions at point C



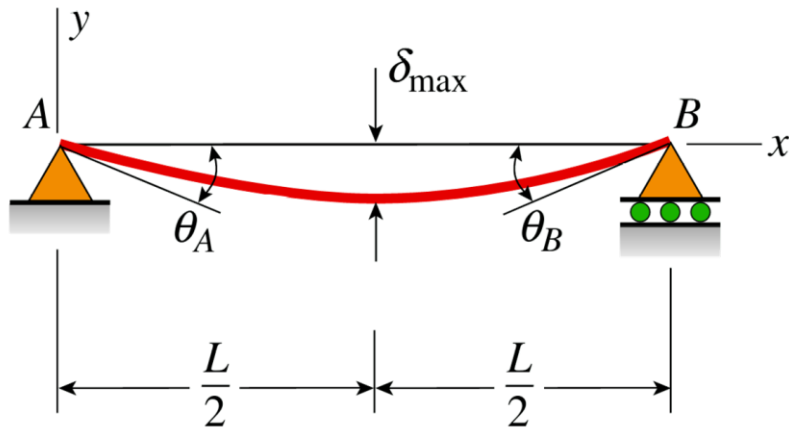
At point C: $(v)_{AC} = (v)_{CB}$
 $(v')_{AC} = (v')_{CB}$



Method of successive integrations



(a)



(b)

So

$$M = -\frac{q}{2}x^2 + \frac{qL}{2}x$$

$$EIv'' = -\frac{q}{2}x^2 + \frac{qL}{2}x$$

$$\Rightarrow EIv' = -\frac{q}{6}x^3 + \frac{qL}{4}x^2 + C_1$$

$$v'\left(\frac{L}{2}\right) = 0 \Rightarrow C_1 = -\frac{qL^3}{24}$$

$$\Rightarrow EIv = -\frac{q}{24}x^4 + \frac{qL}{12}x^3 - \frac{qL^3}{24}x + C_2$$

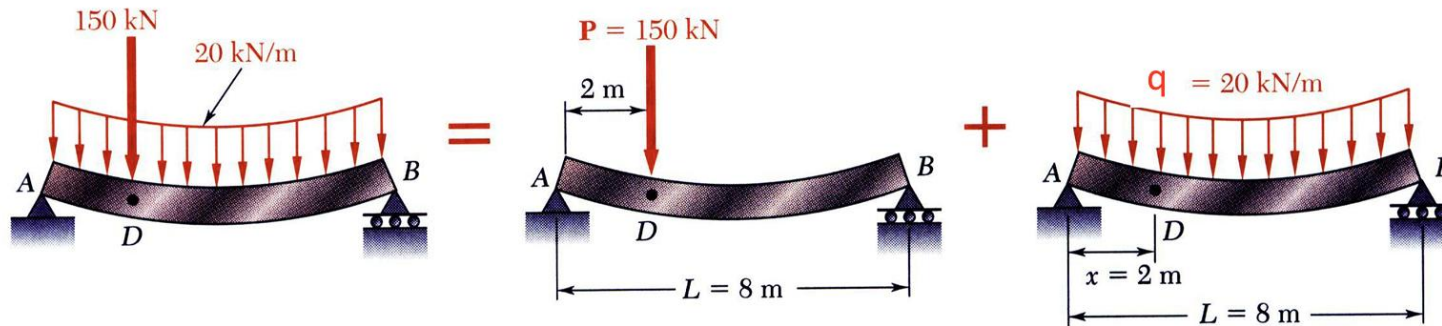
$$v(0) = 0 \Rightarrow C_2 = 0$$

$$\Rightarrow \delta_{\max} = v\left(\frac{L}{2}\right) = -\frac{5qL^4}{384EI}, \quad \theta_A = -v'(0) = \frac{qL^3}{24EI}$$

Method of superposition

Principle of Superposition:

- Deformations of beams subjected to combinations of loadings may be obtained as the linear combination of the deformations from the individual loadings
- Procedure is facilitated by tables of solutions for common types of loadings and supports.



(1) Hooke's law holds for the material

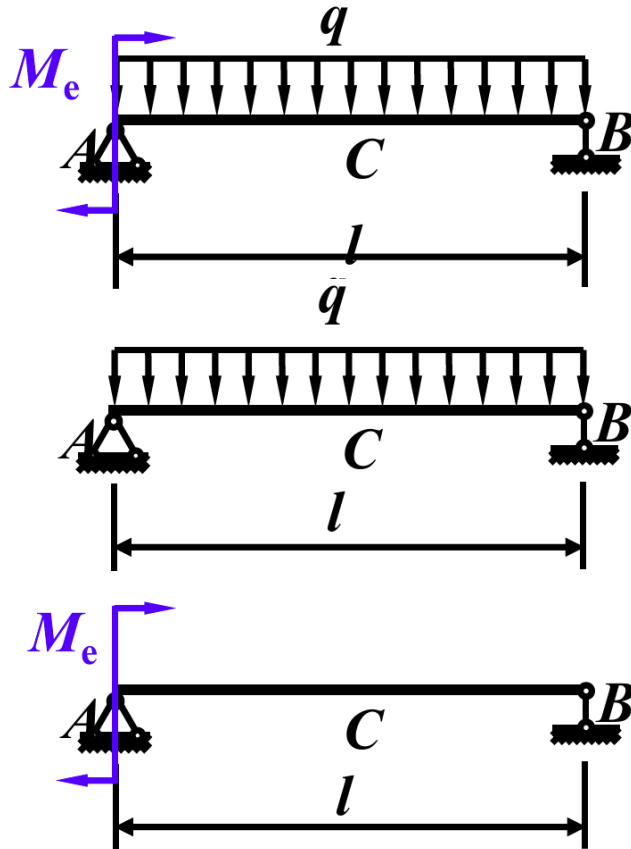
(2) the deflections and rotations are small

(3) the presence of the deflections does not alter the actions of the applied loads.

Example

A simply supported beam has flexural rigidity of EI .

Determine the deflection and deflection angle of central point.



Solution:

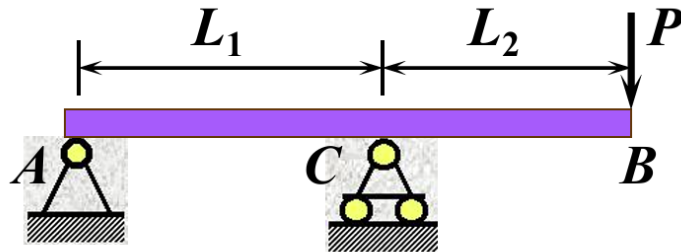
divide the load into two parts

$$w_C = w_{Cq} + w_{CM} = -\frac{5ql^4}{384EI} - \frac{M_e l^2}{16EI}$$

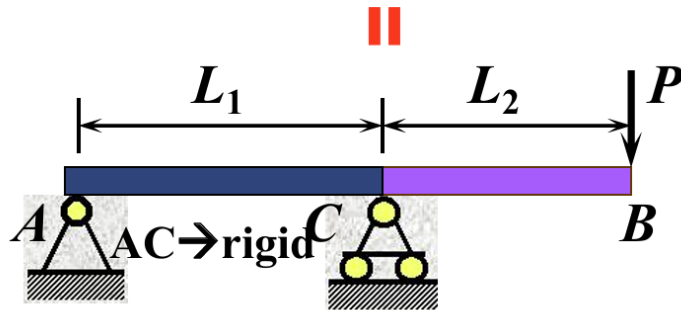
$$\theta_A = \theta_{Aq} + \theta_{AM} = -\frac{ql^3}{24EI} - \frac{M_e l}{3EI}$$

$$\theta_B = \theta_{Bq} + \theta_{BM} = \frac{ql^3}{24EI} + \frac{M_e l}{6EI}$$

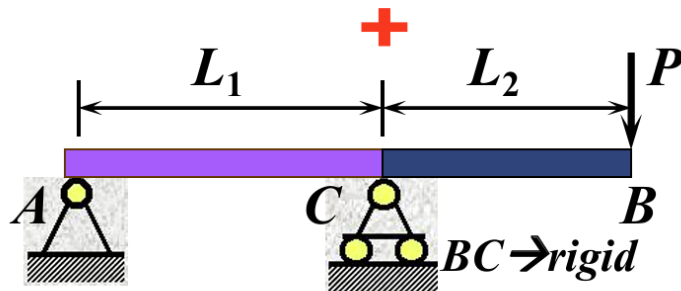
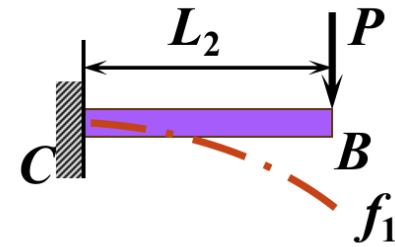
Rigid-body method*



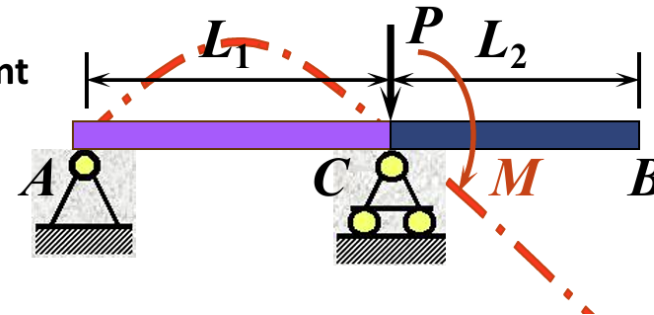
$$f_B = f_1 + f_2$$



Equivalent



Equivalent



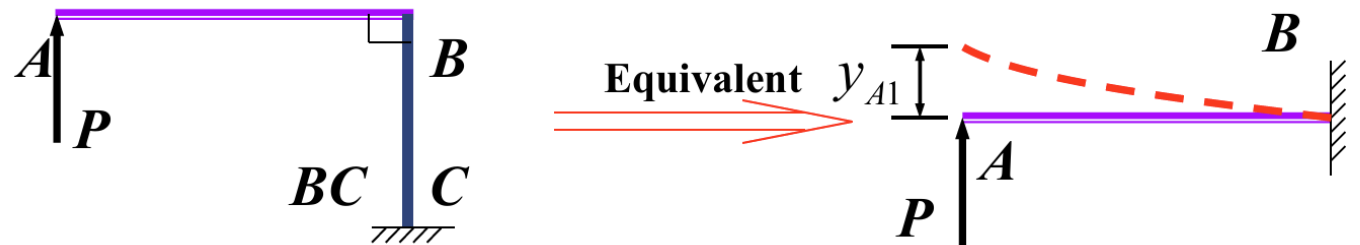
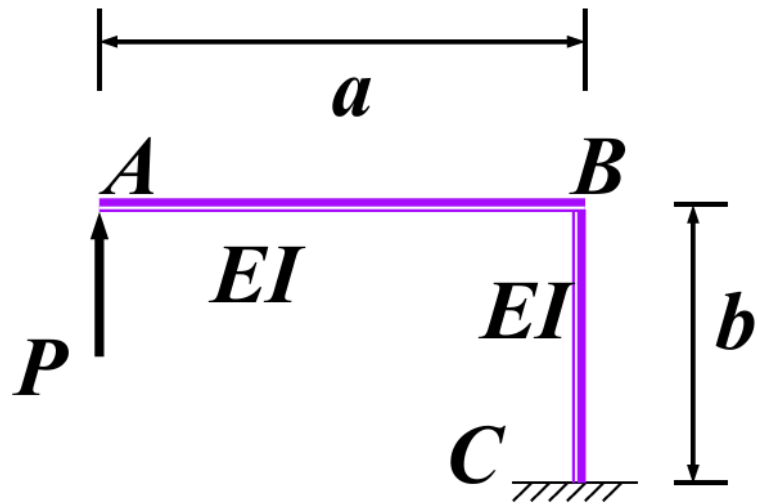
Example*

Determine the horizontal and vertical displacement of free end A of a plane frame (suppose the cross section of AB and BC segments are the same, and ignore the deformation due to axial tension)

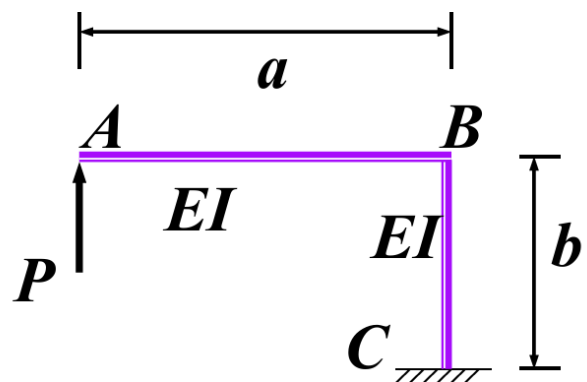
Solution:

1) Make BC as rigid part

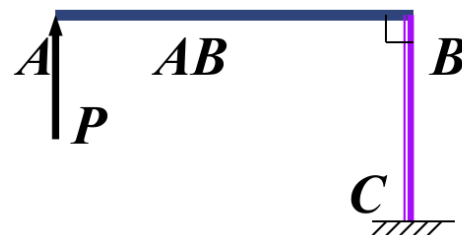
$$x_{A1} = 0 \quad y_{A1} = \frac{Pa^3}{3EI} (\uparrow)$$



Example*

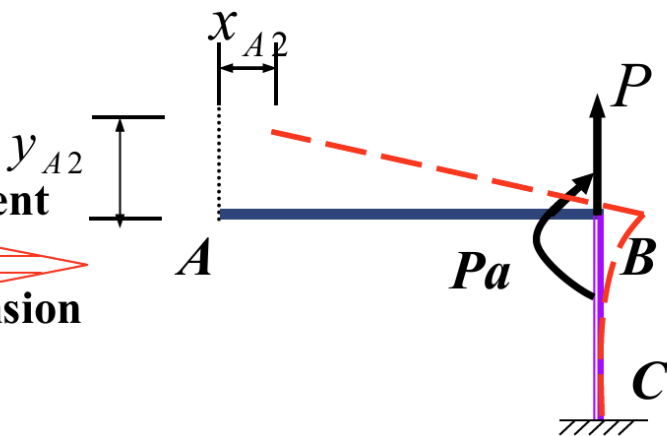


2) Make AB as rigid one



$$x_{A2} = \frac{P a b^2}{2 E I} (\rightarrow)$$

Equivalent
Ignore tension



$$y_{A2} = \theta_B \cdot a = \frac{P a^2 b}{E I} (\uparrow)$$

3) superposition

$$x_A = \frac{P a b^2}{2 E I} (\rightarrow)$$

$$y_A = y_{A1} + y_{A2}$$

$$= \frac{P a^2}{E I} \left(b + \frac{a}{3} \right) (\uparrow)$$

